

Day 13 - Feature Engineering

Feature Engineering Options Considered

The current dataset has only one input feature called feature, which represents the received analog signal value.

I considered a few possible additional features:

1. Distance from the Mean

One idea was to calculate how far each signal value is from the overall or average signal value.

However, this was not useful for the current dataset because the signal values already form two clear and separate groups. Creating another feature based on the distance from the mean would not provide much additional information compared to the original signal value.

2. NTU or Turbidity as a Feature

According to the project requirements, turbidity or NTU information is expected to be used along with the received signal value.

However, the current dataset (1-2NTU.csv) contains data for only one turbidity condition.

Because of this, adding NTU as a feature right now would not be useful because the value would remain the same for all rows.

This feature may become useful later when data from different NTU levels is available.

3. Binning the Signal Values

The feature column contains only 26 unique signal values.

I considered grouping these values into different ranges or bins. However, this was not necessary because the original signal values already contain all the required information.

Therefore, I decided not to use binning.

Raw Feature vs Engineered Features

Based on the previous EDA, the two classes are already clearly separated using the original feature value.

The earlier tests using the fixed threshold method and Logistic Regression also achieved an accuracy of 1.0.

Because the original feature already performs very well, adding artificial features would only increase the complexity without providing any clear improvement.

Therefore, no additional engineered features were added to Dataset V1.

Feature Redundancy and Importance

Since there is only one usable input feature, checking for redundancy between multiple features is not required.

The current feature column provides all the useful information for predicting the output in the available dataset.

Final Feature Used

The feature used for the current model is:

- feature — Received analog signal value

The values are already normalized between 0 and 1.

Feature engineering based on turbidity or NTU will be considered later when data from multiple turbidity conditions becomes available.

At that stage, it will be possible to study the importance of different features and compare their performance across different conditions.